

# Project Risk Management – High-Probability Exam

## Revision Questions

### QUESTION 1

#### Part A

Explain negative risk response strategies.

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#### **Definition**

Negative risk response strategies are methods used to reduce or manage threats that may negatively affect project objectives.

#### **Explanation**

Organizations use different strategies depending on the severity of the risk and the level of authority required to manage it. These strategies help reduce uncertainty, protect project objectives, and improve decision-making.

#### **Key Points**

- Avoid → eliminate the threat completely
- Mitigate → reduce probability or impact
- Transfer → shift risk to another party
- Escalate → move risk to higher management
- Accept → acknowledge risk and prepare contingency plans

#### **Conclusion**

Negative risk response strategies help organizations reduce threats and improve project success.

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## Part B

Give an example to support your answer.

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### **Example**

In the AI-supported governance project, the organization outsourced cybersecurity monitoring to a specialized external vendor to transfer technical security risks and reduce internal exposure.

### **Explanation of Example**

The organization used risk transfer because the external vendor had greater expertise and resources to manage cybersecurity threats effectively.

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## QUESTION 2

## Part A

Differentiate between qualitative and quantitative risk analysis.

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### **Definition**

Qualitative risk analysis prioritizes risks based on subjective assessment, while quantitative risk analysis numerically estimates the impact of risks on project objectives.

### **Explanation**

Qualitative analysis helps identify high-priority risks quickly using probability and impact ratings. Quantitative analysis provides exact numerical estimates related to cost, schedule, and performance impacts.

### **Key Points**

| Qualitative Analysis  | Quantitative Analysis |
|-----------------------|-----------------------|
| Subjective assessment | Numerical assessment  |
| Uses High/Medium/Low  | Uses exact values     |
| Fast and inexpensive  | Detailed and costly   |
| Prioritizes risks     | Measures exact impact |

### **Conclusion**

Qualitative analysis helps prioritize risks, while quantitative analysis provides accurate numerical information for decision-making.

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## **Part B**

Give an example to support your answer.

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### **Example**

In the AI governance project, system integration failure was ranked as a high-priority risk during qualitative analysis, while quantitative analysis estimated a 30% probability of a one-month delay and an 8% increase in project costs.

### **Explanation of Example**

The project team first used qualitative analysis to subjectively classify the system integration failure risk as high probability and high impact. Then, quantitative analysis techniques such as Monte Carlo simulation were used to estimate the numerical effect of the risk on project schedule and operational costs.

## **QUESTION 3**

## **Part A**

Differentiate between risk mitigation and risk avoidance.

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### **Definition**

Risk mitigation reduces the probability or impact of a risk, while risk avoidance eliminates the threat completely.

### **Explanation**

Mitigation is used when the organization wants to reduce the severity of the risk but continue the activity. Since the risk is only reduced and not completely removed, some residual risk usually remains after mitigation.

Avoidance is used when the organization changes the project plan to completely remove the threat. Although avoidance usually eliminates the original risk, the actions taken during mitigation or avoidance may sometimes create secondary risks.

### **Key Points**

| <b>Risk Mitigation</b>        | <b>Risk Avoidance</b>          |
|-------------------------------|--------------------------------|
| Reduces probability or impact | Eliminates the risk completely |
| Risk still exists             | Risk removed                   |
| Minimizes damage              | Prevents occurrence            |

### **Conclusion**

Mitigation reduces the severity of risks, while avoidance completely removes threats from the project.

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## **Part B**

Give an example to support your answer.

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### **Example**

In a software development project, the team added regular system testing to mitigate the risk of software failure, while a risky experimental feature was removed from the project scope to avoid technical problems completely.

### **Explanation of Example**

The project team reduced the possibility of software failure through regular system testing, while technical problems were eliminated by removing the risky experimental feature from the project scope.

## QUESTION 4

### Part A

Explain the project risk management process and its role throughout the project lifecycle.

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#### **Definition**

Project risk management is a continuous process used to identify methodology and risks, analyse (qualitative, quantitative), respond to, implement, and monitor risks throughout the project risk lifecycle in order to reduce threats and increase opportunities.

#### **Explanation**

The project risk management process helps organizations manage uncertainty systematically throughout all phases of the project risk lifecycle. It begins with planning how risks will be managed, followed by identifying risks, analysing their severity, developing response strategies, implementing actions, and continuously monitoring risks as the project progresses from initiation to closure.

#### **Key Points**

- Plan risk management: Define risk management methodology and approach
- Identify threats and opportunities
- Prioritize risks using qualitative analysis
- Estimate numerical impact using quantitative analysis
- Develop response and contingency plans
- Implement planned response actions
- Continuously monitor and review risks

#### **Conclusion**

The project risk management process provides a structured approach for managing uncertainty throughout the project lifecycle, reducing threats, increasing opportunities, and improving overall project success.

## Part B

Give an example to support your answer.

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### **Example**

In a software development project, the team first prepared a Risk Management Plan that defined the risk management methodology, they then identified risks related to system failure and schedule delays. The risks were prioritized using qualitative analysis, while quantitative analysis estimated possible project delays and additional costs. The team later planned risk responses such as backup servers and testing procedures before implementing the responses and continuously monitoring risks throughout the project lifecycle.

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### **Explanation of Example**

The project followed the risk management process throughout the project risk lifecycle by identifying risks, analysing their impact, planning responses, implementing actions, and continuously monitoring risks from project initiation until project completion.

## QUESTION 5

## Part A

Explain the core concept of risk management at the project, program, portfolio, and organizational levels.

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### **Definition**

Risk management is applied differently across organizational levels depending on the objectives, scope, and type of value being managed.

### **Explanation**

At the project level, risk management focuses on achieving project objectives while controlling operational and tactical risks related to scope, cost, schedule, and performance.

At the program level, risk management focuses on optimizing benefits and coordinating risks across multiple related projects and activities.

At the portfolio level, risk management focuses on achieving organizational strategic objectives, balancing opportunities and threats, and optimizing risk-return trade-offs.

At the organizational or enterprise level, risk management focuses on strategic alignment, governance, resilience, and long-term organizational success.

### **Key Points**

| Level                       | Core Concept                                            |
|-----------------------------|---------------------------------------------------------|
| Project                     | Deliver project objectives and manage operational risks |
| Program                     | Optimize benefits across related projects               |
| Portfolio                   | Achieve strategic objectives through value delivery     |
| Organizational / Enterprise | Strategic alignment and enterprise-wide risk governance |

### **Important**

- Project risks are mainly tactical and operational
- Program risks focus on integrated benefits
- Portfolio risks focus on strategic value realization
- Organizational risks focus on enterprise-wide uncertainty and governance

### **Conclusion**

Risk management evolves across organizational levels from operational project execution to strategic organizational governance and enterprise-wide value delivery.

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## **Part B**

Give an example to support your answer.

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### **Example**

A software issue initially affected only one project and was managed by the project team. Later, the issue began affecting multiple connected projects, so program management coordinated the response to protect overall program benefits. As the impact expanded further, portfolio and organizational management evaluated how the issue affected strategic objectives, organizational priorities, and enterprise operations.

### **Explanation of Example**

The same risk was managed differently at each organizational level depending on the scope of impact and the organizational objectives affected.

## **QUESTION 6**

### **Part A**

Explain positive risk response strategies.

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### **Definition**

Positive risk response strategies are methods used to increase or manage opportunities that may positively affect project objectives.

### **Explanation**

Organizations use different opportunity response strategies depending on the potential benefits and organizational objectives affected. These strategies help maximize opportunities, improve value delivery, and increase the likelihood of project success.

### **Key Points**

- Exploit → ensure the opportunity happens
- Enhance → increase probability or positive impact
- Share → partner with another party to maximize benefits
- Escalate → move opportunity to higher management
- Accept → acknowledge the opportunity and benefit if it occurs

### **Conclusion**

Positive risk response strategies help organizations maximize opportunities, improve value delivery, and support project success.



## Part B

Give an example to support your answer.

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### **Example**

In a software development project, the team discovered a new automated testing tool that could significantly reduce testing time and improve software quality. To ensure the organization fully benefited from this opportunity, management decided to officially adopt the tool, allocate additional budget for its implementation, and require the development team to use it throughout the testing process.

### **Explanation of Example**

The project team used the exploitation strategy because the organization invested additional resources and officially adopted the new testing tool to ensure the opportunity would occur. By fully integrating the tool into the project, the organization improved project performance, reduced testing time, and maximized software quality.